

Alexandre Binninger

✉ alexandre.binninger@inf.ethz.ch

in binninga

🐙 AlexandreBinninger

🌐 alexandrebinninger.com

✂ @TheGraphicsFrog



Work Experience

- 2021 – current **Interactive Geomtry Lab (ETH Zurich)**, Scientific assistant / Doctoral Student
- 2023 (4 months) **Meta**, Research scientist internship
Integrated the Gemini research group.
Supervised by Dr. Philipp Herholz.
- 2019 (6 months) **Programming Methodology Group (ETH Zurich)**, Research intern
Study of the Completeness, Termination and Efficiency for e-matching-based Axiomatisations.
Supervised by Dr. Alexander J. Summers.
- 2018 (3 months) **Muvee Technologies**, Software engineering internship.
Integrated a development team working on an automatic photobook-maker application.
- 2016 – 2017 **Gendarmerie Nationale**, Leadership training as Cadet Officer
École Polytechnique's first-year military service. Optimized duty hours across police units.

Education

- 2021 – current **Ph.D. in Computer Science, ETH Zurich** Computer Graphics
Research interests: generative AI, 3D representations, shape modeling.
Advised by Prof. Dr. Olga Sorkine-Hornung.
- 2019 – 2021 **M.Sc. Computer Science**, ETH Zurich.
Focus on visual computing courses.
Thesis: *Approximation of 3D Shapes by Developable Surfaces via Tangent Space Thinning*.
- 2016 – 2019 **Master of Engineering, Science and Technology**, École Polytechnique.
Algorithms and foundations of programming languages track.
- 2017 – 2018 **Bachelor of Philosophy**, Université Paris-Nanterre.
Distance learning alongside scientific studies. Awarded with honors.
- 2014 – 2016 **Preparatory Program** Lycée Louis-Le-Grand.
A two-year post secondary intensive coursework leading to competitive entrance examinations to the *Grandes Écoles*. Main subjects: Mathematics and Physics.

Research Publications

- 1 **A. Binninger**, R. Wiersma, P. Herholz, and O. Sorkine-Hornung, "TetWeave: Isosurface Extraction using On-The-Fly Delaunay Tetrahedral Grids for Gradient-Based Mesh Optimization," *ACM Trans. Graph.*, vol. 44, no. 4, Aug. 2025, *SIGGRAPH 2025 issue*.
- 2 **A. Binninger** and O. Sorkine-Hornung, "SD- π XL: Generating Low-Resolution Quantized Imagery via Score Distillation," in *SIGGRAPH Asia 2024 Conference Papers*, ser. SA '24, Tokyo, Japan: Association for Computing Machinery, Dec. 2024, ISBN: 9798400711312.

- 3 **A. Binninger**, A. Hertz, O. Sorkine-Hornung, D. Cohen-Or, and R. Giryes, “SENS: Part-Aware Sketch-based Implicit Neural Shape Modeling,” *Computer Graphics Forum*, vol. 43, no. 2, e15015, Apr. 2024, *Honorable Mention*.
- 4 **A. Binninger** and O. Sorkine-Hornung, “Smooth Interpolating Curves with Local Control and Monotone Alternating Curvature,” *Computer Graphics Forum (proceedings of SGP 2022)*, vol. 41, no. 5, Jul. 2022, ISSN: 1467-8659.
- 5 **A. Binninger***, F. Verhoeven*, P. Herholz, and O. Sorkine-Hornung, “Developable Approximation via Gauss Image Thinning,” *Computer Graphics Forum (proceedings of SGP 2021)*, vol. 40, no. 5, pp. 289–300, Jul. 2021, **Equal contribution*, ISSN: 1467-8659.

Skills

Languages	French (native), English (proficient), German (intermediate), Mandarin Chinese (intermediate).
Programming	Python, C++, JavaScript, Rust.
AI & Graphics Libraries	PyTorch, libigl, OpenGL, ThreeJS.
Web Development	HTML, CSS, Node.js, React, Tailwind CSS, MySQL.
Design & Visualization	Blender, Adobe Illustrator, Inkscape.
Technical Tools	LaTeX, Git, Unix, Docker.
Research & Professional	Academic research, scientific writing, project management, teaching, public speaking.

Outside Activities

Miscellaneous Experience

- 2024 – 2025 **Talent Kick**, TK-S24 batch
Talent Kick is an incubator dedicated to students enrolled in Swiss Universities. Co-founded Fabricaible, a start-up providing generative models to create fabricable designs. Pitched at venture pitching event and in front of the Talent Kick strategy board.
- 2023 – 2025 **VMI board member**, ETH D-INFK scientific staff association
VMI represents, supports and connects the scientific staff of the computer science department of ETH. Organized many events, such as 3 career-focused panel discussions with a focus on academic excellence. Co-president between Oct. 2024 and Feb. 2025.
- 2021 – 2025 **Teaching Assistant**, ETH Zurich
Taught 2 semesters in Linear Algebra, 3 semesters in Shape Modeling courses. One semester as Head TA for Linear Algebra. My role was to manage other teaching assistants, create graded homework, and design the final exam.
- 2019 – 2020 **Discord community - Developer and Leader**, E-sport Community
Created, developed and managed a Discord server reaching 1'000 members. Tournament Organizer. Designed and implemented a Discord bot hosted on a remote VPS featuring a ranking system, a match wizard, an automated competitive event manager and a profile card generator with gamified features. Implementation with discord.py, Python Imaging Library (PIL) and MySQL.

Outside Activities (continued)

2017 – 2018

- **Team Research Project**, Live theater subtitling Project
Displayed subtitles in real time on smart glasses or a smartphone with a web interface.
Designed the algorithm matching the actors' spoken text with their written text.
Winners of the Vivatech Competition. Project presented at the international Microsoft Imagine Cup.
- **Binet Réseau**, Computer Technology association at École Polytechnique
Deployed a web service for equipment pool among students on the campus.
Integrated the Sysadmin team. Provided support for students' computer-related problems.

Hobbies

- Sports ■ Calisthenics, volleyball, hiking.
- Cultural ■ Philosophy, self-help books, video games, cooking.